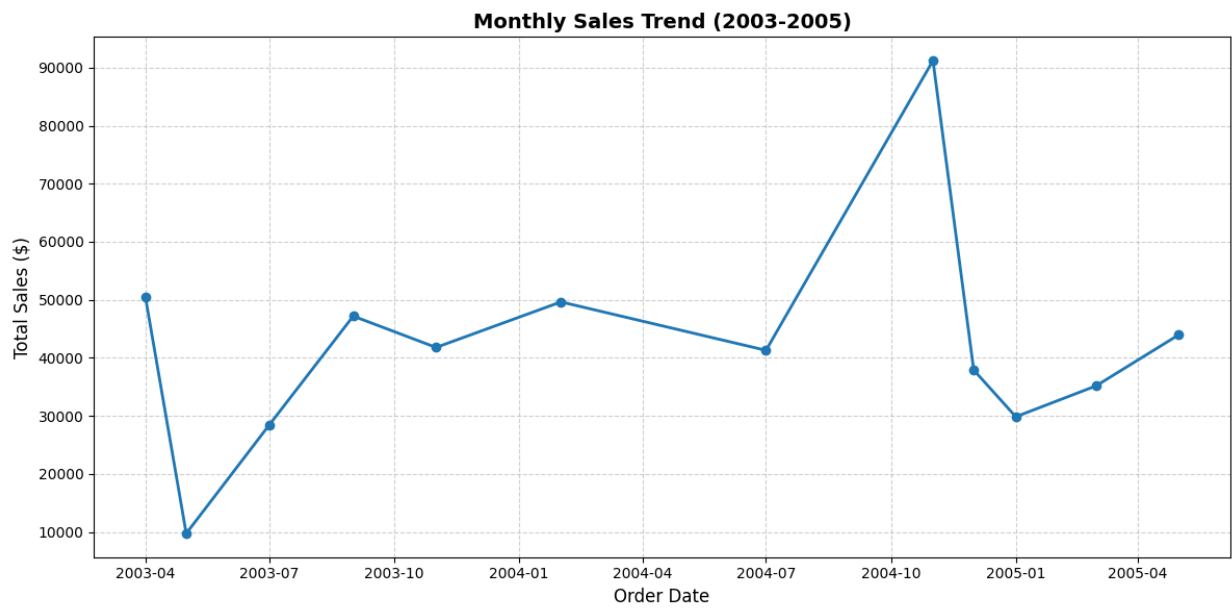


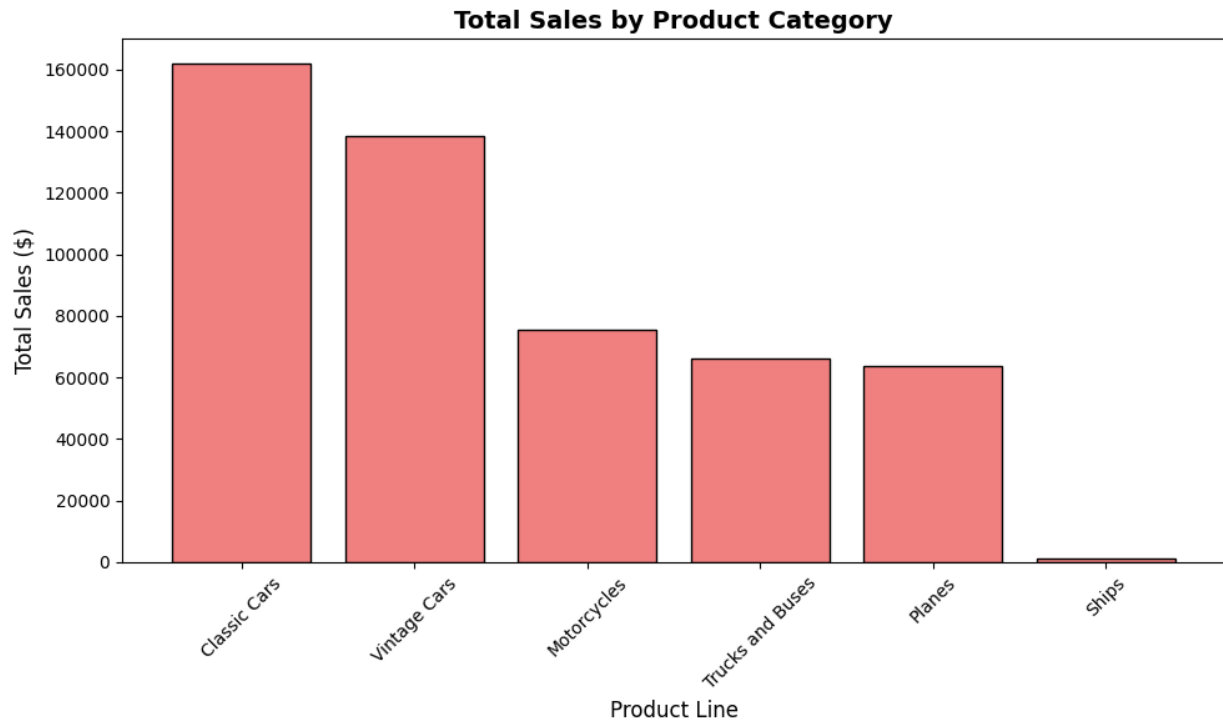
Visualization Summary and Design Choices

To effectively analyze and communicate sales patterns, multiple chart types were selected based on the nature of the data and the insights required.

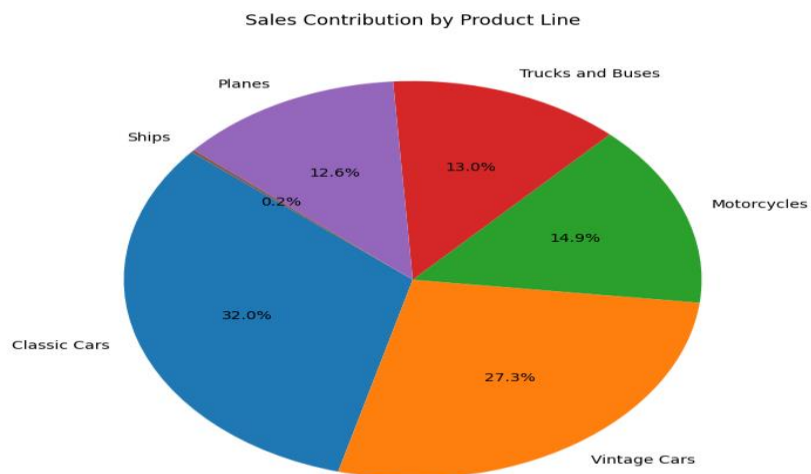
A **line chart** was used to visualize the monthly sales trend because it clearly shows changes in sales over time and helps identify peaks, declines, and overall patterns. The x-axis represents the order date, while the y-axis shows total sales, both clearly labeled for readability. Grid lines and markers were added to improve trend visibility, and an appropriate figure size ensured clarity.



A **bar chart** was chosen to compare total sales across different product categories. Bar charts are ideal for categorical comparisons, making it easy to identify which product lines generate the highest revenue. Axis labels, rotated x-axis tick labels, and consistent color styling were applied to enhance readability and prevent overlap.



A **pie chart** was used to represent the percentage contribution of each product line to total sales. This chart effectively highlights the relative share of each category. Percentage labels were included using autopct, and a clear title was added to ensure quick interpretation.



Across all visualizations, meaningful titles, axis labels, and consistent formatting were applied to improve interpretability and ensure the charts are presentation-ready.